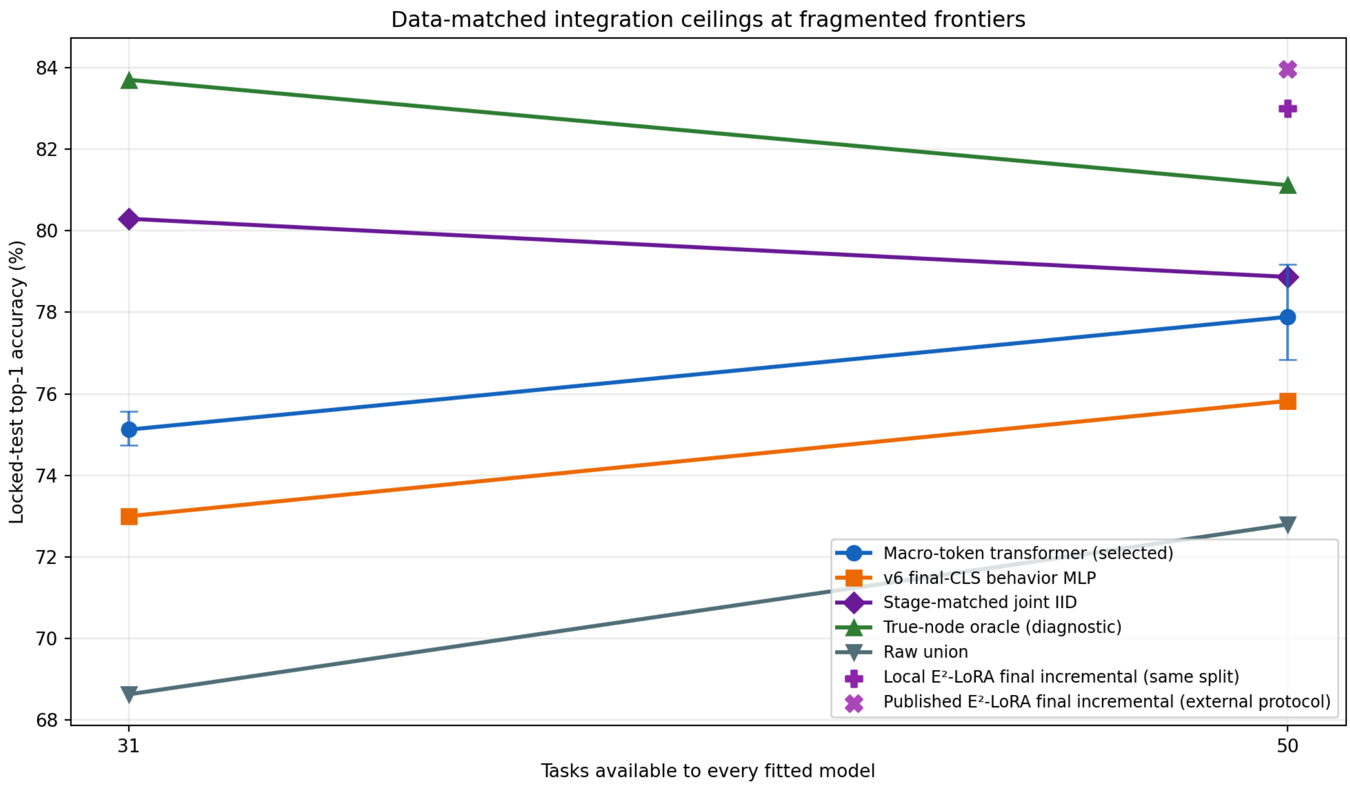
# ImageNet-R-50 Macro-Token Integrator

Clean architecture selection, all-training refit, and locked-test ceiling study

> **Selected model.** 1 transformer block, learning rate 0.0003, chosen by mean clean validation NLL. The run has no accuracy gate.

**Main result.** Relative to the data-matched v6 MLP, macro-token accuracy changed by +2.123 points at stage 31 and +2.061 points at stage 50. Its signed differences from stage-matched joint IID were -5.171 and -0.978 points.

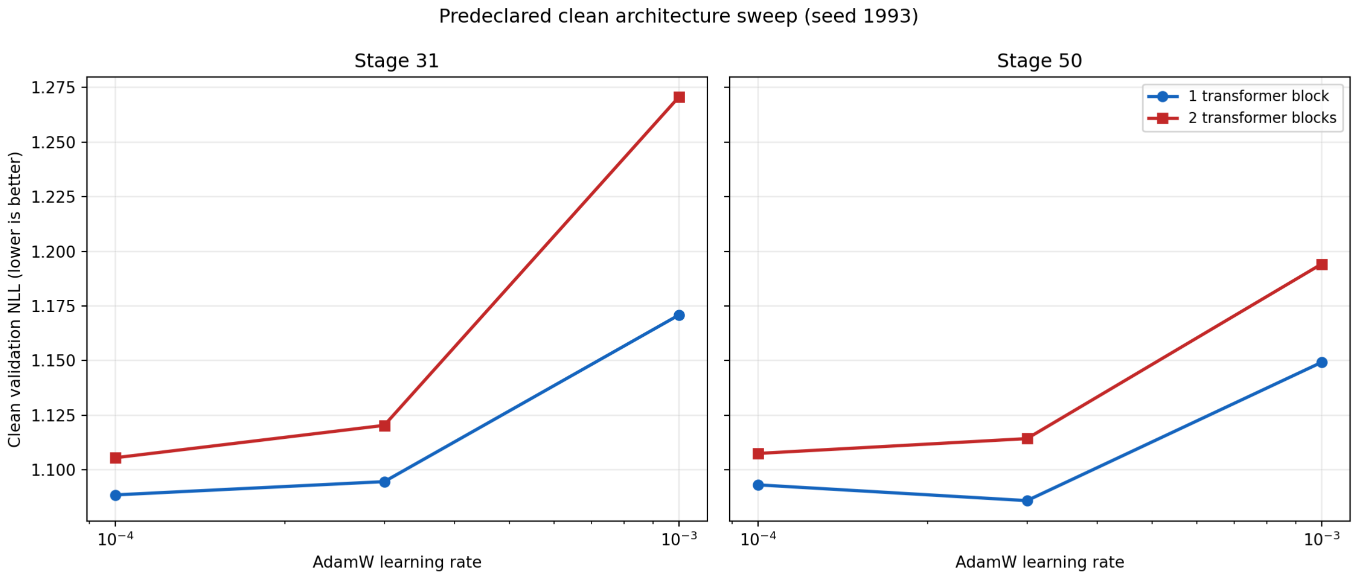
## Primary result




Names here and in every table are identical. Macro-token and v6 points are three-seed means; error bars show the observed macro-token range.

| Stage | Macro-token transformer (selected) | v6 final-CLS behavior MLP | Raw union | True-node oracle (diagnostic) | Stage-matched joint IID | Local $E^2$-LoRA final incremental (same split) |
|---|---|---|---|---|---|---|
| 31 | 75.122% | 73.000% | 68.632% | 83.700% | 80.294% | — |
| 50 | 77.889% | 75.828% | 72.800% | 81.117% | 78.867% | 82.995% |

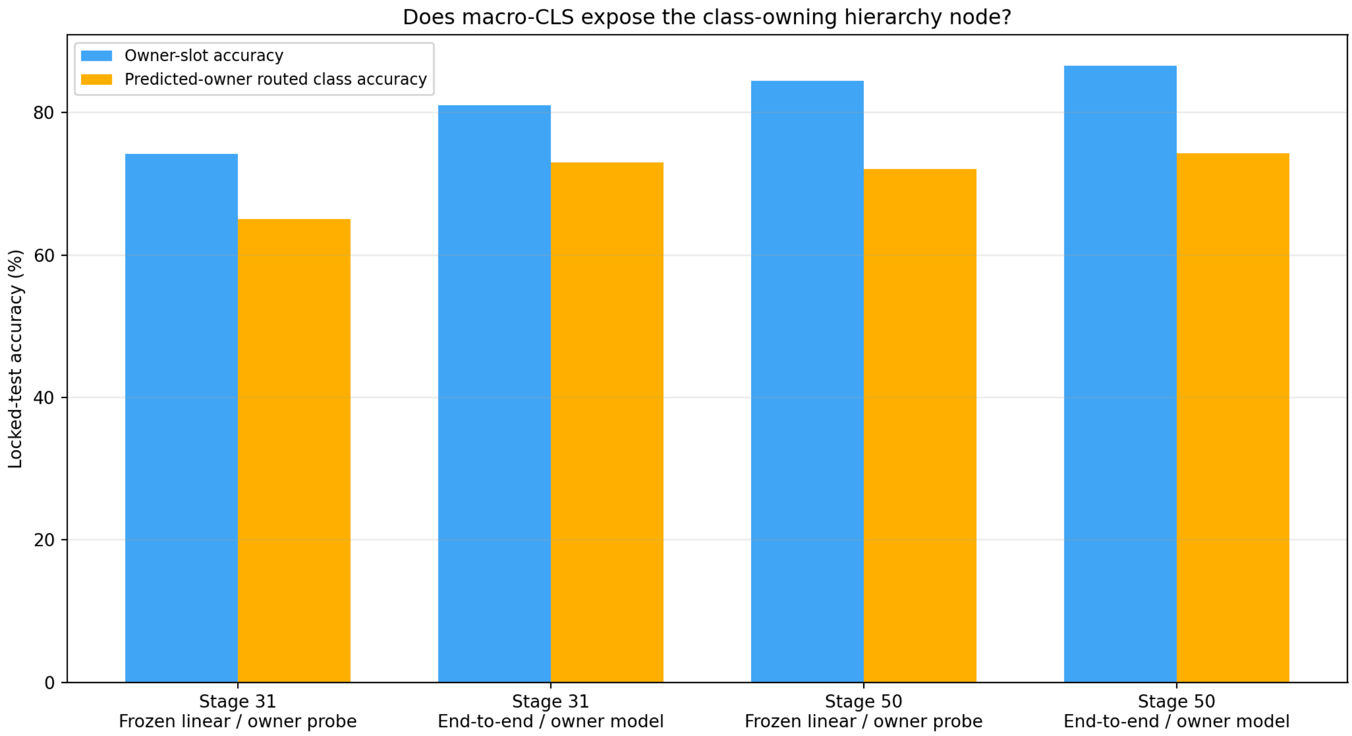
# Clean selection




Only training-derived clean validation chose depth and learning rate. Rejected cells were not evaluated on test.

Each node contributes all 197 final adapted ViT tokens. Corresponding fixed-slot positions are projected from 4,608 to 768, a compact 3,606-value behavior META token is appended, and one or two width-768 transformer blocks classify directly from macro-CLS. No raw-union residual skip is present.

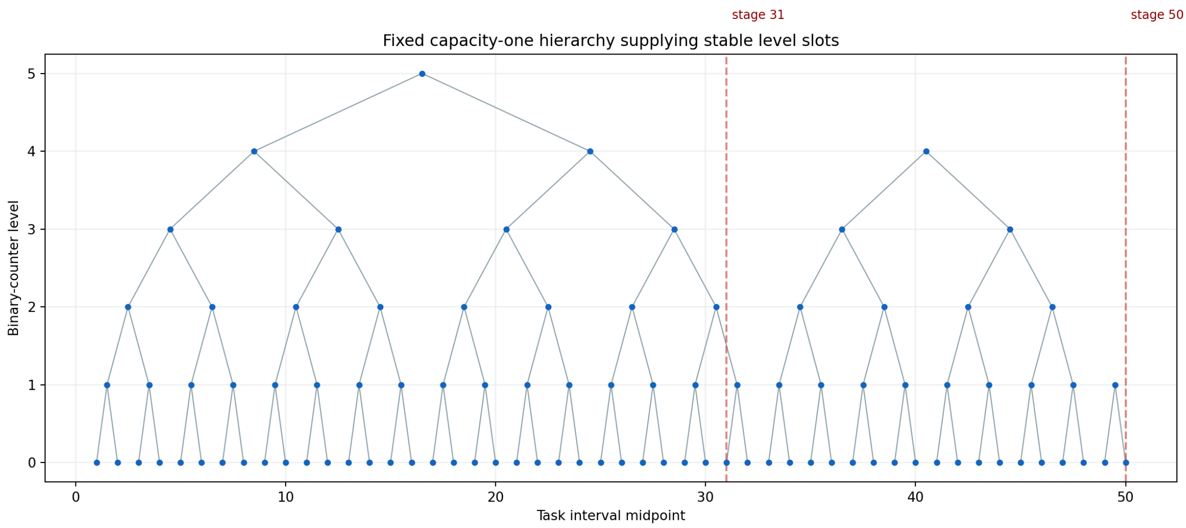
# Owner diagnostics




The frozen linear probe measures linearly available owner information in class-trained macro-CLS. The separate end-to-end model optimizes the same inputs for owner prediction. Neither receives labels at inference.

| Stage | Partition | Diagnostic | Owner accuracy | Routed class accuracy |
|---|---|---|---|---|
| 31 | fit | Frozen linear owner probe | 81.466% | 78.506% |
| 31 | fit | End-to-end owner model | 90.093% | 87.240% |
| 31 | validation | Frozen linear owner probe | 74.221% | 63.431% |
| 31 | validation | End-to-end owner model | 79.501% | 69.498% |
| 50 | fit | Frozen linear owner probe | 89.266% | 85.891% |
| 50 | fit | End-to-end owner model | 94.120% | 91.161% |
| 50 | validation | Frozen linear owner probe | 81.000% | 67.208% |
| 50 | validation | End-to-end owner model | 85.125% | 71.958% |
| 31 | test | Frozen linear owner probe | 74.161% | 65.068% |
| 31 | test | End-to-end owner model | 81.027% | 72.956% |
| 50 | test | Frozen linear owner probe | 84.450% | 72.067% |
| 50 | test | End-to-end owner model | 86.600% | 74.300% |

# Fixed hierarchy




Stage 31 has five live level slots and stage 50 has three. Clean and all-training hierarchies share this topology but use different upstream training populations.

Stage-matched joint IID is a fresh rank-16 QKV-plus-fc1 LoRA and affine head trained jointly on each exact prefix for five epochs. True-node oracle is label-aware and diagnostic. The v6 MLP receives the same cached node computation and training populations as the macro model, isolating the information boundary and head structure.

These are head-only dense multiply-accumulate estimates, not measured latency. Every condition also performs the listed number of node-specific adapted ViT forwards.

| Stage | Condition | Active ViT forwards/image | Head dense MACs/image |
|---|---|---|---|
| 31 | Macro-token transformer (selected) | 5 | 2,043,889,152 |
| 31 | v6 final-CLS behavior MLP | 5 | 9,117,696 |
| 50 | Macro-token transformer (selected) | 3 | 1,811,498,496 |
| 50 | v6 final-CLS behavior MLP | 3 | 9,117,696 |

The clean hierarchy excluded all 4,800 validation identities from upstream training. Every all-training refit completed before the training seal, which recorded zero test-token requests. BF16 token shards were capped at 64 GiB and removed after sealing. Immediate replay performed zero hierarchy optimizer work and zero adapted-token forwards.